EWB AWS INTERNSHIP PROJECT REPORT

PROJECT TITLE: AWS VPC Peering: Connect Two VPCs and Prove Private Connectivity

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1.ABSTRACT

This project focuses on establishing secure private communication between two Amazon Virtual Private Clouds (VPCs) using AWS VPC Peering. Two VPCs with non-overlapping IPv4 CIDR blocks were configured with separate subnets to host private application instances. An EC2 bastion host was deployed in the public subnet of VPC A to provide administrative SSH access to the private instances.

A VPC peering connection was established between the two VPCs, and the required routes were configured in the route tables to enable communication between the private workloads. Security group inbound rules were configured to control SSH and ICMP traffic between the designated hosts. Connectivity was validated through forward and reverse ping tests using private IP addresses, along with SSH access to App-B through the bastion host. A controlled ICMP failure and recovery test was also included to demonstrate the role of security group rules in network communication.

The project demonstrates the practical application of VPC networking, subnetting, route tables, security groups, bastion-based administration, and private connectivity in Amazon Web Services (AWS). It also highlights the importance of correct routing, return paths, and security rules when troubleshooting connectivity between separate VPCs.

Keywords: Amazon Web Services (AWS), Amazon VPC, VPC Peering, EC2, Route Tables, Security Groups, Bastion Host, Private Connectivity.

2.INTRODUCTION

Amazon Virtual Private Cloud (Amazon VPC) enables users to create isolated virtual networks within AWS and control their IP address ranges, subnets, routing, and network access. By default, resources in separate VPCs cannot communicate using their private IP addresses. VPC Peering provides a networking connection between two VPCs, allowing resources in each VPC to communicate using private IP addresses, provided that the required routes and security rules are configured correctly.

In this project, two VPCs were created in the US East (N. Virginia) AWS Region. VPC-A uses the CIDR block 10.10.0.0/16, while VPC-B uses 10.20.0.0/16. An EC2 bastion host was deployed in a public subnet of VPC-A, and two EC2 instances, App-A and App-B, were deployed in private subnets across the two VPCs.

A VPC Peering connection was established between VPC-A and VPC-B. Route tables and security group rules were configured to enable private communication between the application instances. Connectivity was verified using SSH and ICMP ping tests.

The project demonstrates how VPC Peering can provide private communication between resources in separate AWS virtual networks without requiring the application instances to have public IP addresses.

3. AWS SERVICES AND TECHNOLOGIES USED:

- The project uses the following AWS services and technologies to establish and validate private communication between two Virtual Private Clouds (VPCs).
- Amazon Virtual Private Cloud (VPC): Provides isolated virtual networks for the two application environments.
- VPC Peering: Connects VPC A (10.10.0.0/16) and VPC B (10.20.0.0/16) to enable private IPv4 communication.
- Amazon EC2: Hosts the bastion server and two private application instances.

- Route Tables: Direct traffic within each VPC and across the peering connection using the appropriate destination CIDRs.
- Security Groups: Control inbound SSH and ICMP traffic to the lab instances.
- Internet Gateway: Provides internet connectivity to the public subnet containing the bastion host.
- Amazon Linux 2023 and OpenSSH: Provide the operating environment and secure remote access to the instances.
- ICMP and ping: Validate connectivity and demonstrate controlled network failure and recovery.

4. ARCHITECTURE AND SYSTEM DESIGN

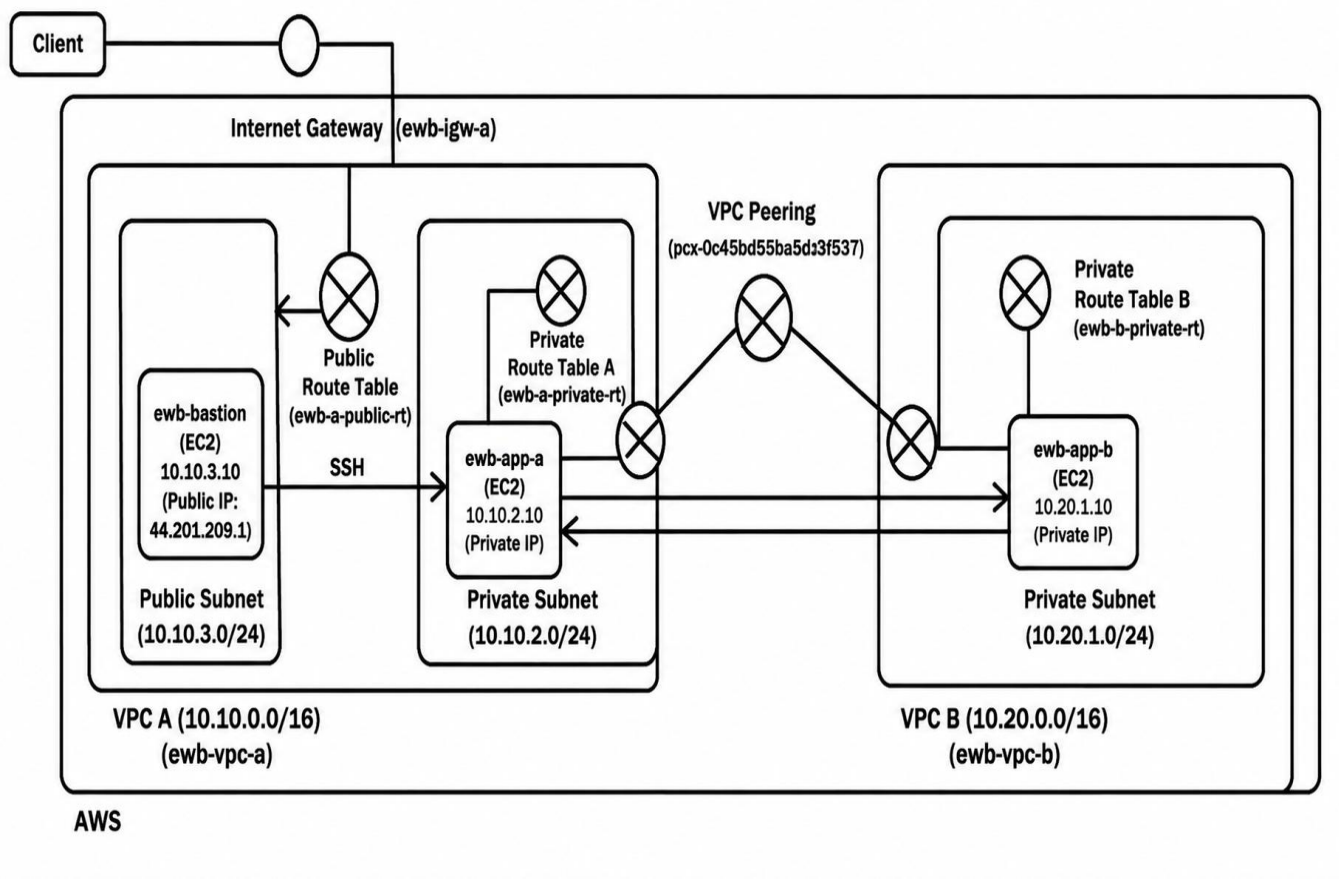


Figure 1: AWS VPC Peering Architecture for Private Communication Between Two VPCs

The project consists of two isolated VPCs connected through an active VPC peering connection in the AWS us-east-1 Region. VPC A uses the CIDR block 10.10.0.0/16, while VPC B uses 10.20.0.0/16. Since the CIDR blocks do not overlap, the two VPCs can communicate through peering when the required routes and security rules are configured.

VPC A contains a public subnet hosting the bastion instance and a private subnet hosting App-A. VPC B contains a private subnet hosting App-B. Both application instances use private IPv4 addresses and do not require public IP addresses.

The bastion host provides an administrative entry point to the private instances. Route tables on both sides direct traffic destined for the peer VPC through the peering connection. Security groups restrict access to the required SSH and ICMP traffic.

Forward path: App-A sends traffic to App-B's private IP address (10.20.1.10). The route table in VPC A directs the traffic through the peering connection to VPC B.

Return path: App-B sends its response to App-A's private IP address (10.10.2.10). The route table in VPC B directs the response back through the same peering connection.

5.AWS RESOURCE INVENTORY:

The following table summarizes the AWS resources used to implement and validate the VPC peering architecture in the us-east-1 Region.

Resource	Name	Configuration / Identifier
VPC A	ewb-vpc-a	10.10.0.0/16
VPC B	ewb-vpc-b	10.20.0.0/16
Public subnet	ewb-a-public-subnet	10.10.1.0/24
Additional public subnet	ewb-a-public-subnet-1a	10.10.3.0/24
Private subnet A	ewb-a-private-subnet	10.10.2.0/24
Additional private subnet A	ewb-a-private-subnet-1a	10.10.4.0/24

Private subnet B	ewb-b-private-subnet	10.20.1.0/24
Bastion instance	ewb-bastion	Private IP: 10.10.3.10
Application instance A	ewb-app-a	Private IP: 10.10.2.10
Application instance B	ewb-app-b	Private IP: 10.20.1.10
Internet gateway	ewb-igw-a	Attached to VPC A
Peering connection	ewb-vpc-a-to-vpc-b	pcx-0c45bd55ba5d3f537
Public route table	ewb-rt-a-public	Public subnet routing
Private route table A	ewb-rt-a-private	Local and VPC B peering routes
Private route table B	ewb-rt-b-private	Local and VPC A peering routes
Bastion security group	ewb-sg-bastion	SSH access from the permitted client IP
App-A security group	ewb-sg-app-a	SSH and ICMP access as configured
App-B security group	ewb-sg-app-b	SSH and ICMP access as configured

The peering connection was active during testing. Both application instances used private IP addresses for communication. The additional subnets in VPC A were created during troubleshooting and are included here for completeness.

6.SUBNET CONFIGURATION AND ROUTE TABLES

6.1 Subnet Configuration

The project uses IPv4 CIDR blocks to divide the two VPCs into separate subnets. VPC A contains public and private subnets, while VPC B contains a private subnet for App-B

Subnet	VPC	IPv4 CIDR	Purpose
ewb-a-public-subnet	VPC A	10.10.1.0/24	Hosts public-facing administration access
ewb-a-public-subnet-1a	VPC A	10.10.3.0/24	Additional public subnet containing the bastion
ewb-a-private-subnet	VPC A	10.10.2.0/24	Hosts App-A
ewb-a-private-subnet-1a	VPC A	10.10.4.0/24	Additional private subnet created during troubleshooting
ewb-b-private-subnet	VPC B	10.20.1.0/24	Hosts App-B

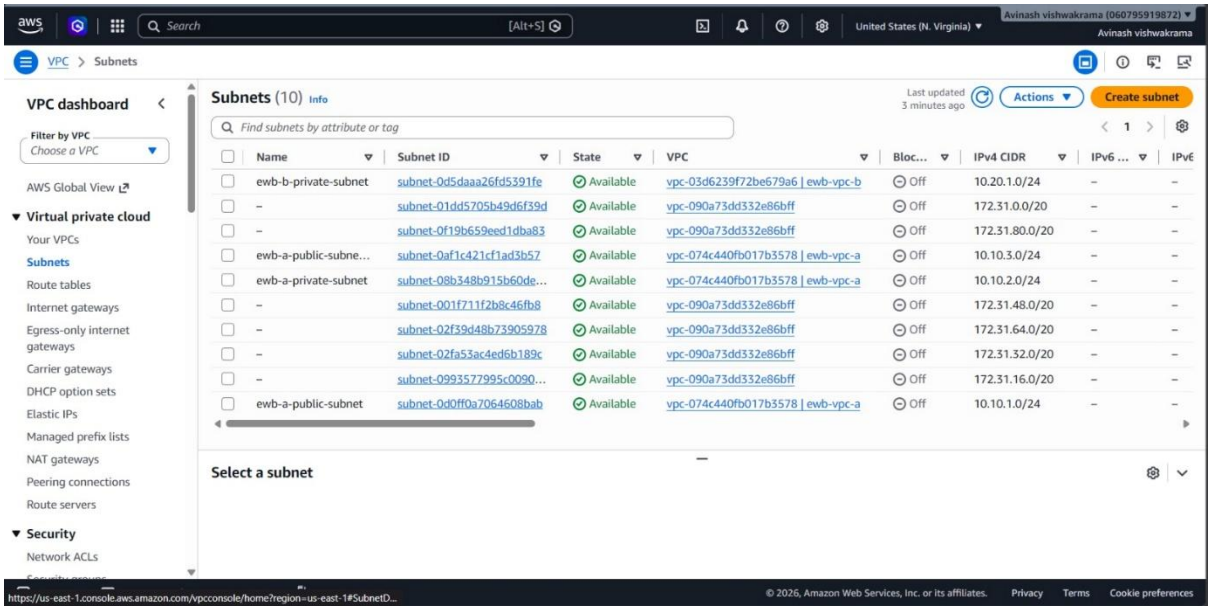


Figure 2: Overview of the project subnets and their IPv4 CIDR blocks.

6.2 Route Table Configuration

Three custom route tables are used to direct traffic within the VPCs and across the peering connection.

Destination	**Target**	**Purpose**
10.10.0.0/16	local	Routes traffic within VPC A
0.0.0.0/0	ewb-igw-a	Provides internet routing for the public subnets
10.20.0.0/16	pcx-0c45bd55ba5d3f537	Routes traffic destined for VPC B through peering

Route table 1: ewb-rt-a-public

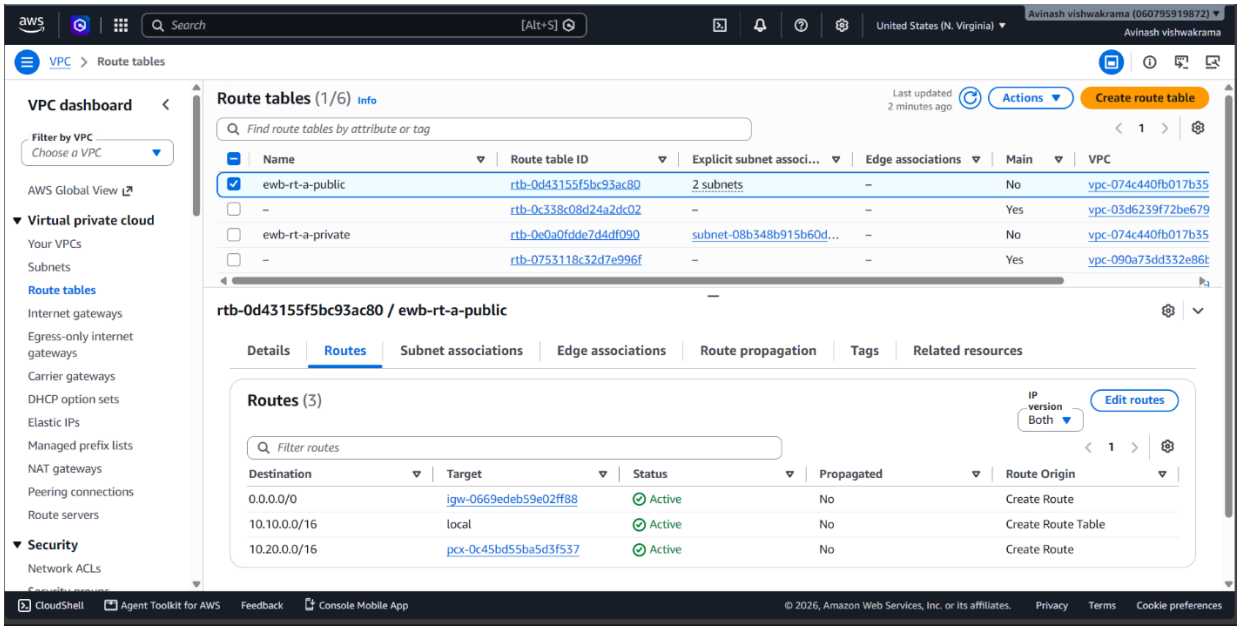


Figure 3: Public route table showing local, Internet Gateway, and VPC peering routes

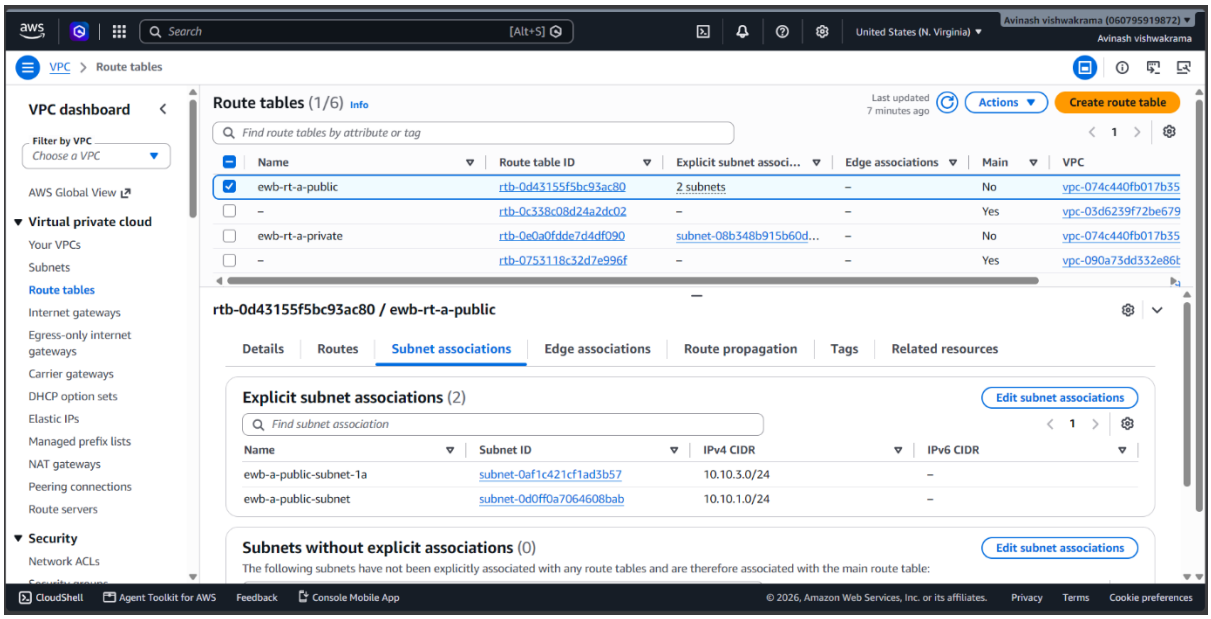


Figure 4: Subnet associations for the public route table in VPC A.

Destination	**Target**	**Purpose**
10.10.0.0/16	local	Routes traffic within VPC A
10.20.0.0/16	pcx-0c45bd55ba5d3f537	Routes traffic from VPC A to VPC B

Route table 2: ewb-rt-a-private

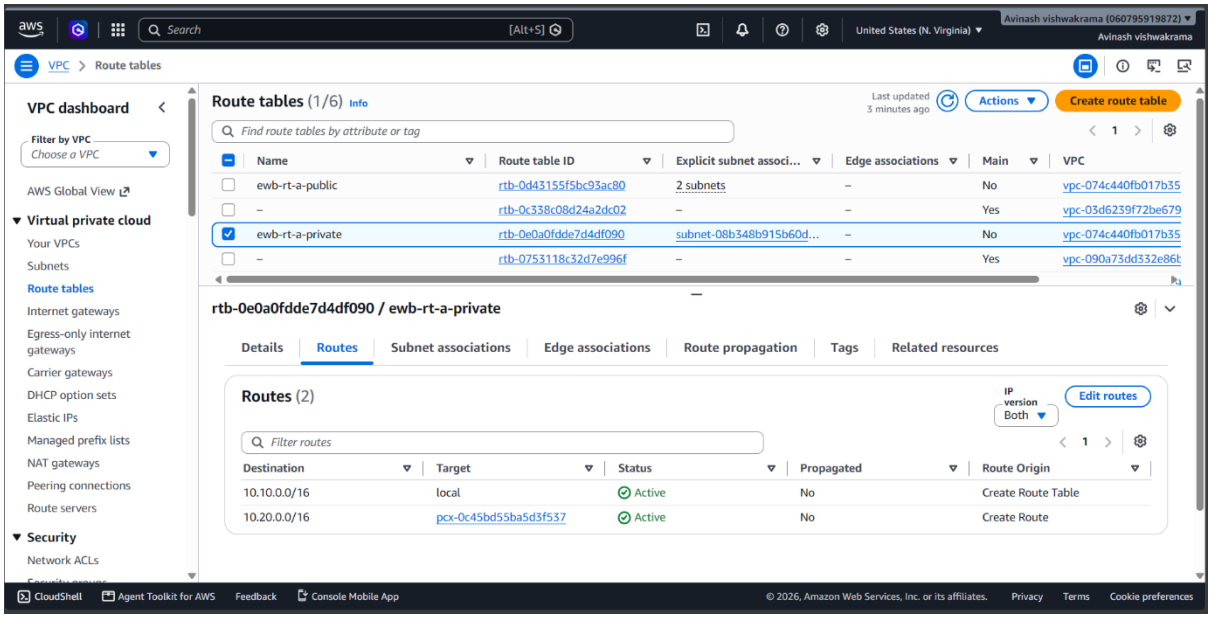


Figure 5: Private route table in VPC A showing the route to VPC B

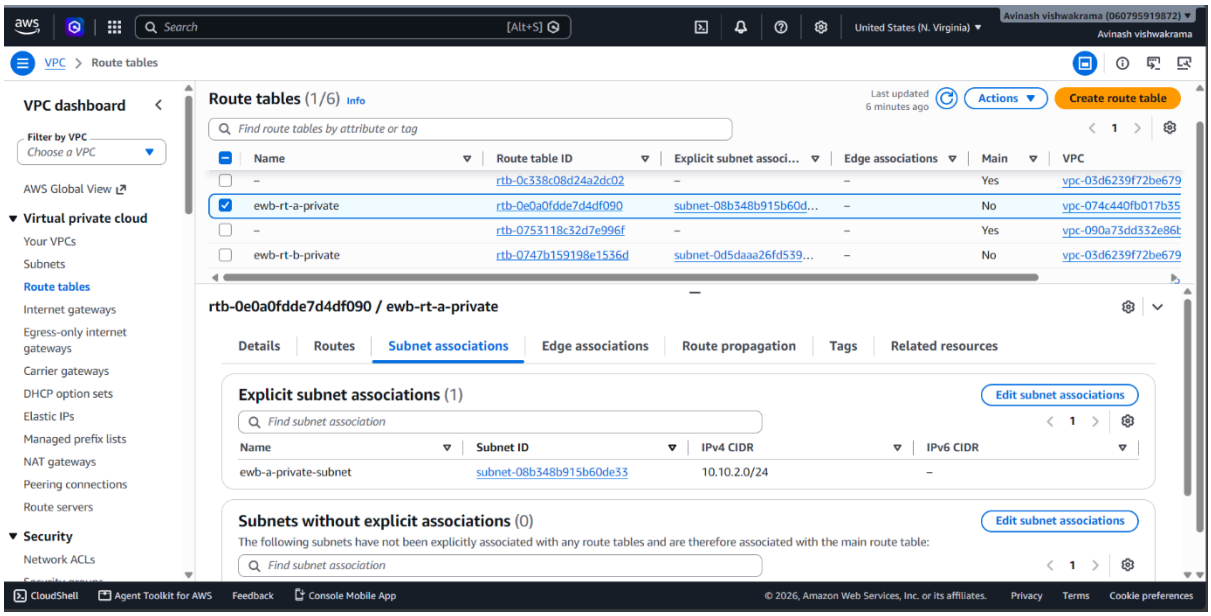


Figure 6: Subnet association for the private route table in VPC A.

Destination	Target	Purpose
10.20.0.0/16	local	Routes traffic within VPC B
10.10.0.0/16	pcx-0c45bd55ba5d3f537	Routes return traffic from VPC B to VPC A

Route table 3: ewb-rt-b-private

6.3 Routing Explanation

VPC peering provides a private network path between the two VPCs, but routes must be configured on both sides. The route in VPC A directs packets for 10.20.0.0/16 to the peering connection, while the route in VPC B directs packets for 10.10.0.0/16 to the same connection.

The public route table also contains the Internet Gateway route for the bastion's administrative connectivity. The private application instances do not require public IP addresses to communicate with one another through VPC peering.

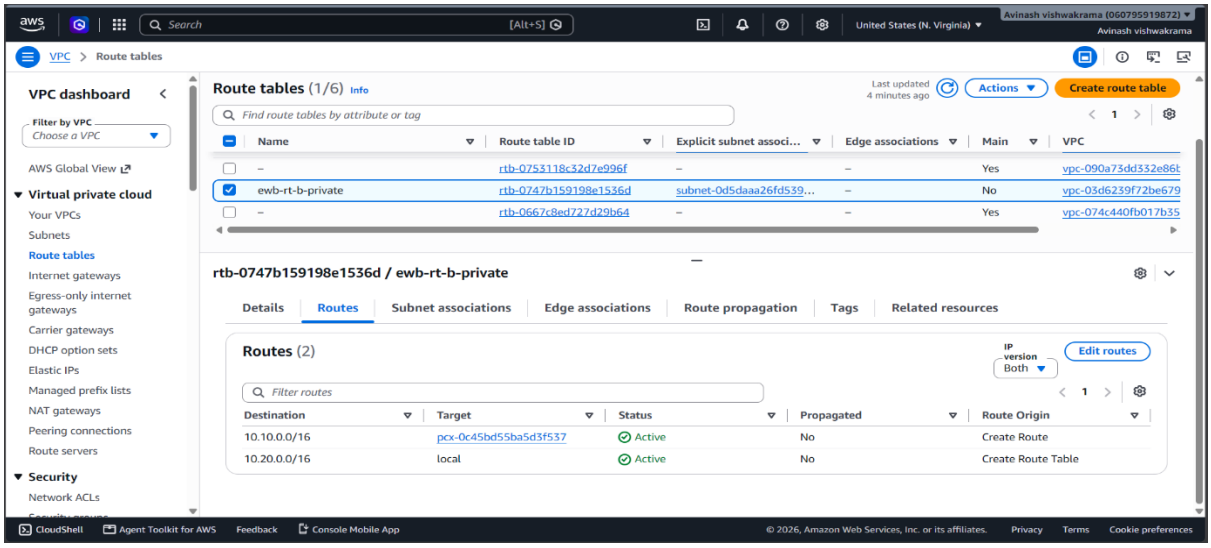


Figure 7: Private route table in VPC B showing the route to VPC A through the peering connection.

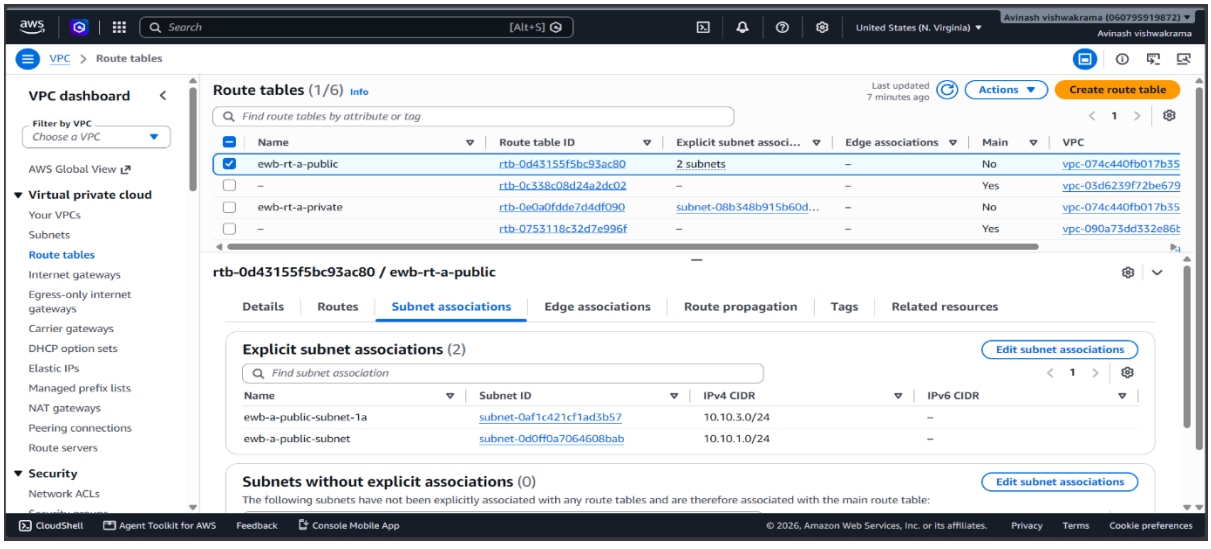


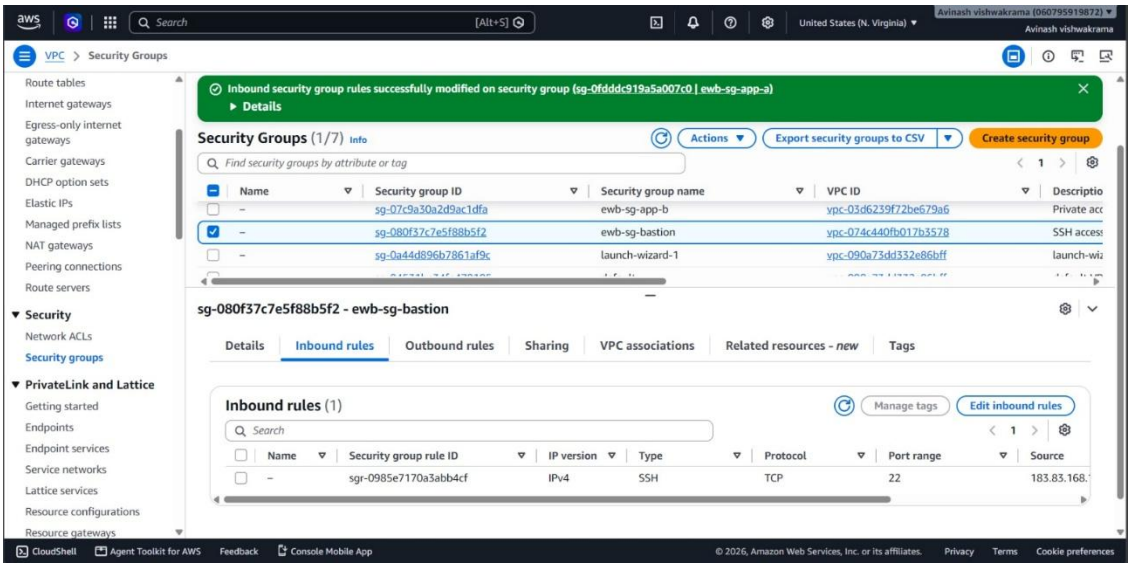
Figure 8: Subnet association for the private route table in VPC B.

7. SECURITY GROUPS AND ACCESS CONTROL

Security groups were configured to control traffic to the bastion host and the two private application instances. SSH was used for secure remote administration, while ICMP was used to verify private connectivity between the application instances.

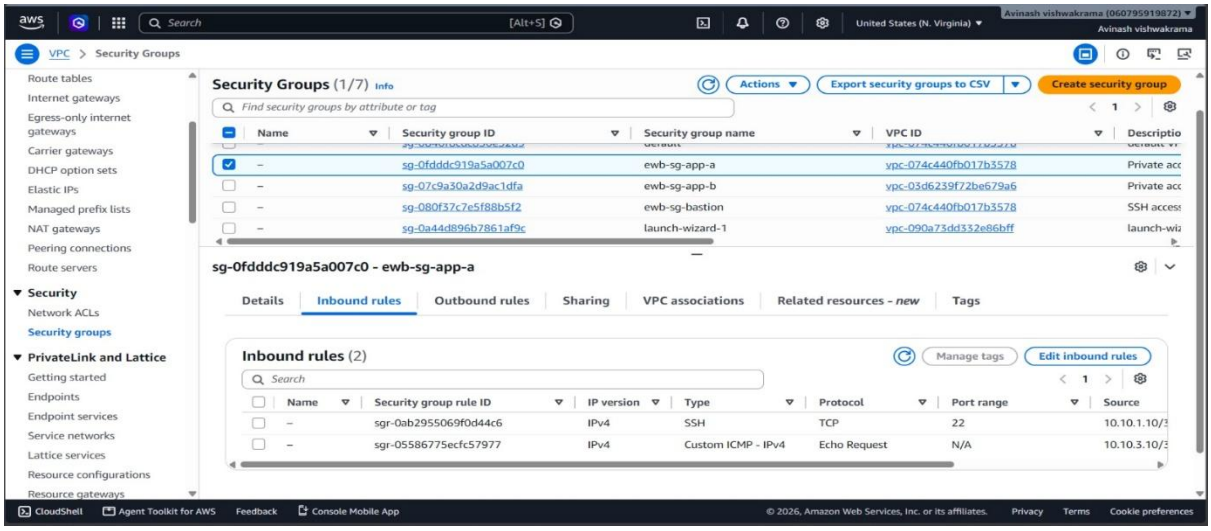
7.1 Bastion Security Group;

The ewb-sg-bastion security group controls access to the bastion instance in VPC A. Its SSH inbound rule allows administrative access from the permitted client IP address. The bastion host acts as the entry point for connecting to the private application instances.



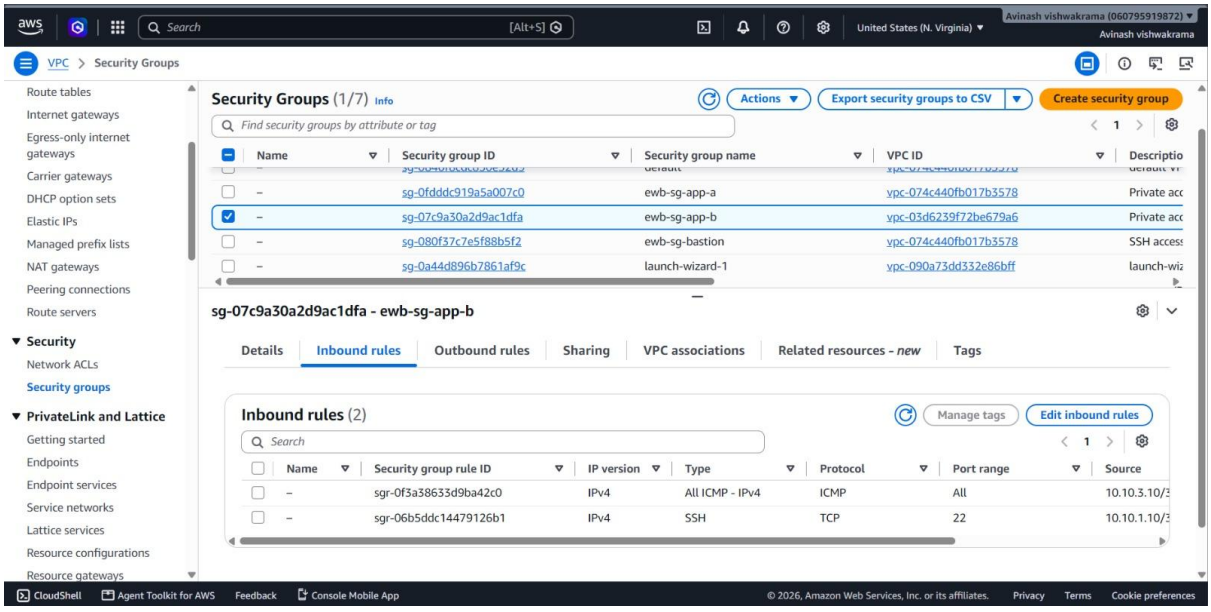
7.2 App-A Security Group:

The ewb-sg-app-a security group protects App-A, which uses private IP address 10.10.2.10. Its inbound rules permit the required SSH and ICMP traffic from the configured sources.



7.3 App-B Security Group:

The ewb-sg-app-b security group protects App-B, which uses private IP address 10.20.1.10. Its inbound rules permit SSH access from the configured source and ICMP traffic for connectivity testing. The ICMP rule was temporarily removed and restored to demonstrate controlled failure and recovery.



7.4 Security Validation:

The security-group configuration was validated through SSH access and ping tests. The bastion host successfully provided access to both private instances. After the ICMP rule was removed from App-B's security group, the ping test from App-A failed. Restoring the rule allowed the ping test to succeed again.

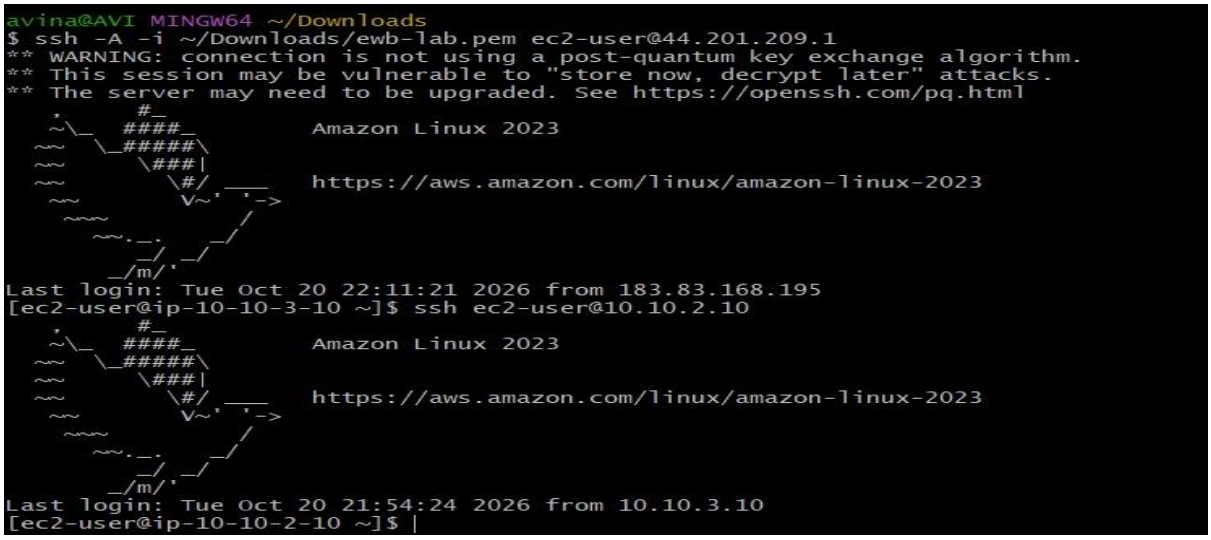


Figure 12: Successful SSH connection from the bastion host to App-A

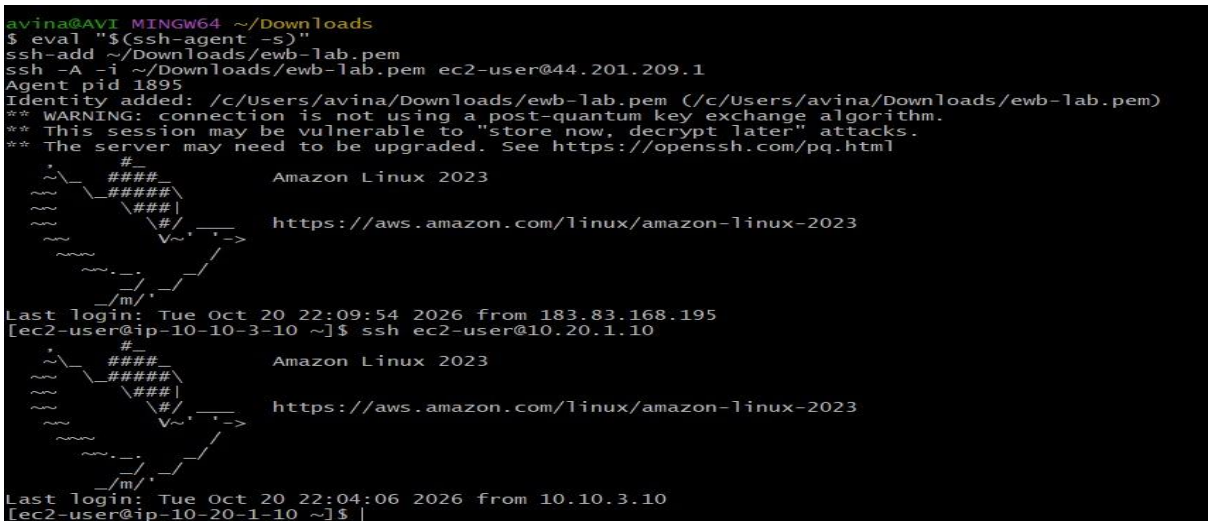


Figure 13: Successful SSH connection from the bastion host to App-B

8. IMPLEMENTATION AND CONNECTIVITY TESTING

The VPC peering connection was tested using SSH and ICMP traffic between the EC2 instances. The bastion host provided administrative access to the private instances, while the configured routes enabled communication between VPC A and VPC B through the private IP addresses.

8.1 SSH Connectivity

SSH connectivity was verified from the bastion host to App-A (10.10.2.10) and App-B (10.20.1.10). Both connections were successful, confirming that the configured security-group rules permitted SSH access from the bastion host.

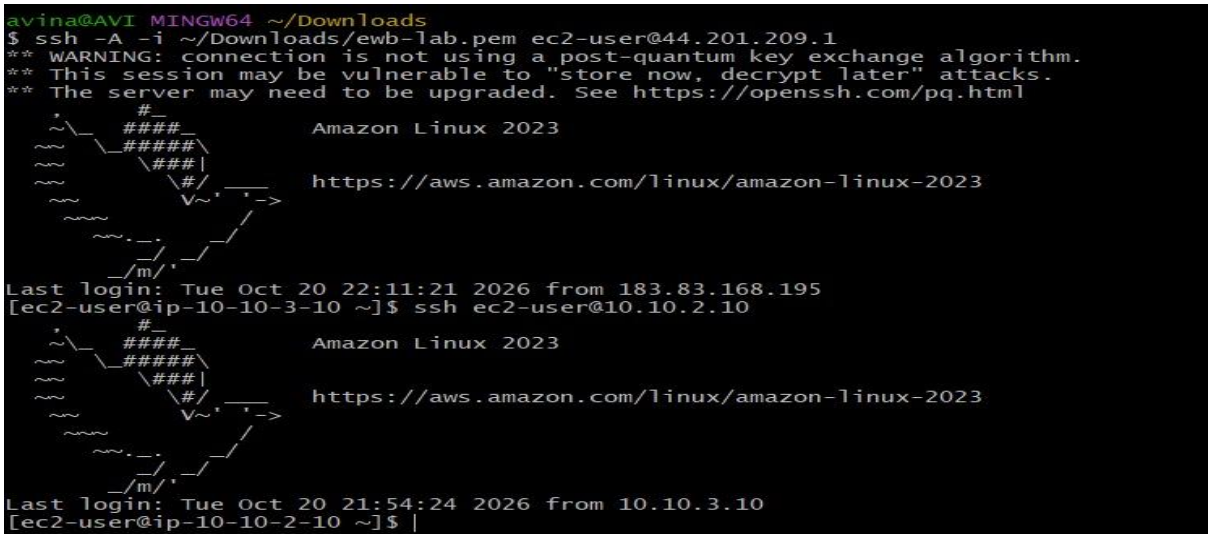


Figure 14: Successful SSH connection from the bastion host to App-A.

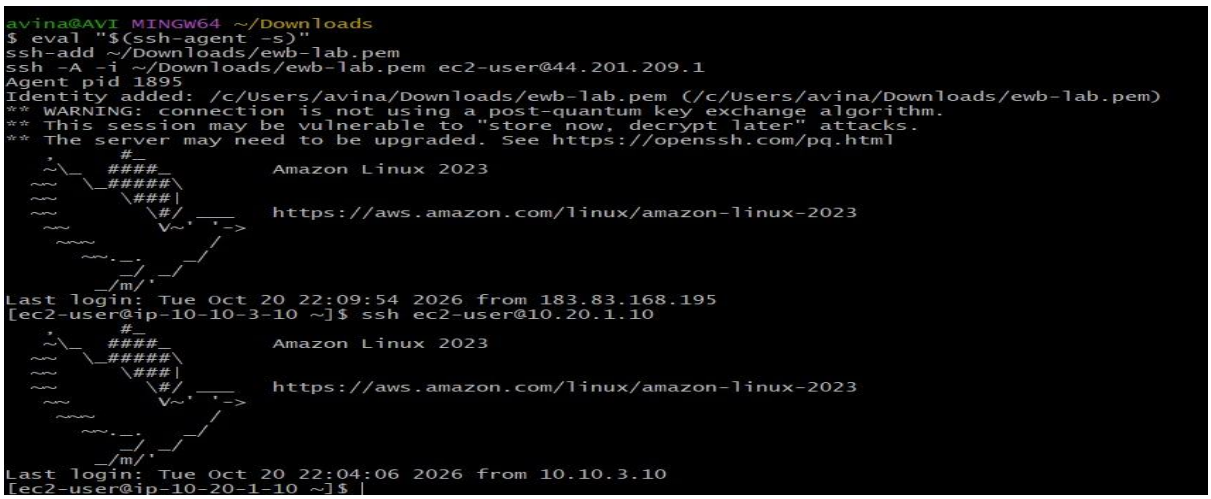


Figure 15: Successful SSH connection from the bastion host to App-B.

8.2 ICMP Connectivity: App-A to App-B

A ping test was executed from App-A (10.10.2.10) to App-B (10.20.1.10). Four ICMP packets were transmitted and received, with 0% packet loss, confirming successful private connectivity between the two VPCs.

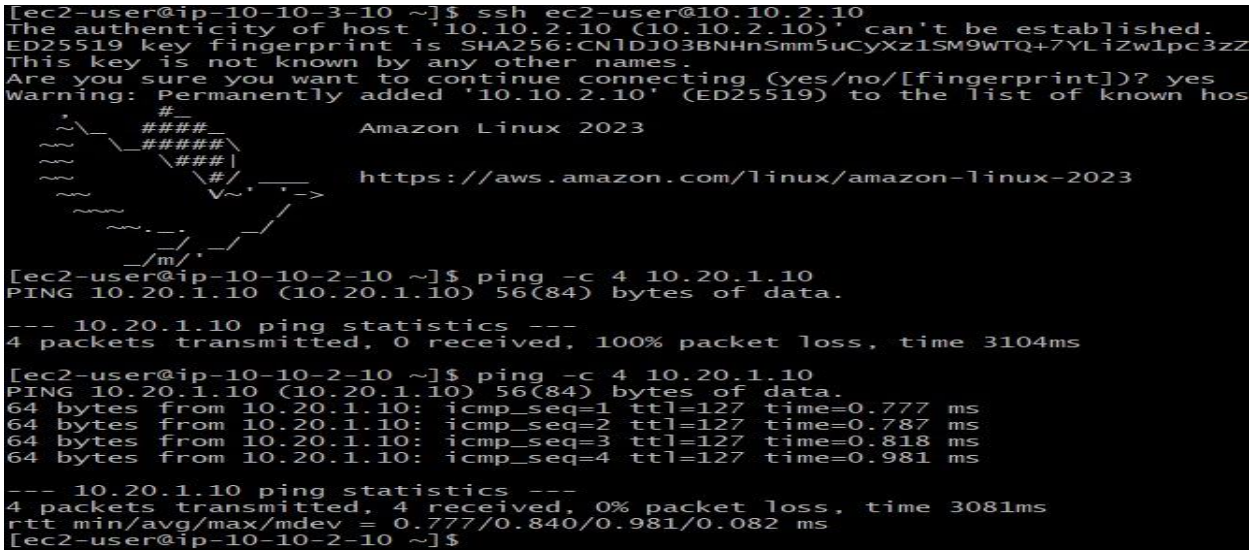


Figure 16: Successful ICMP connectivity from App-A to App-B.

8.3 ICMP Connectivity: App-B to App-A

A reverse ping test was executed from App-B (10.20.1.10) to App-A (10.10.2.10). Four ICMP packets were transmitted and received, with 0% packet loss. This verified bidirectional private communication through the VPC peering connection.

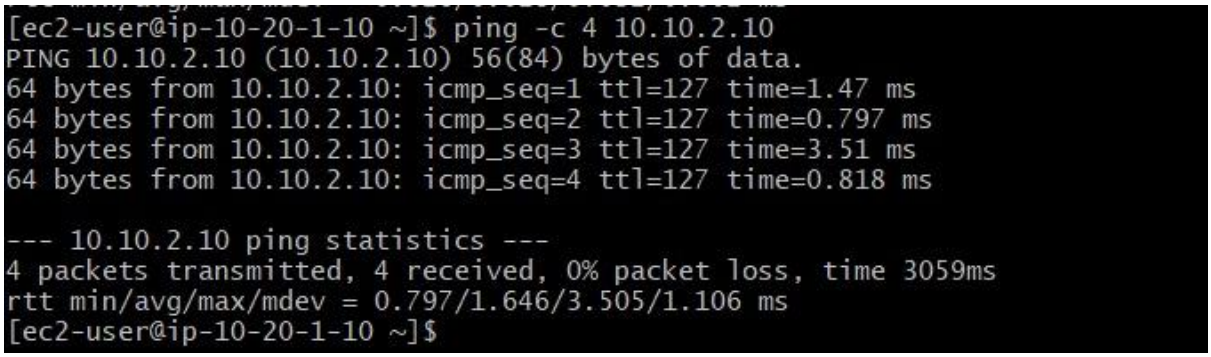


Figure 17: Successful ICMP connectivity from App-B to App-A.

8.4 Test Results

The connectivity tests confirmed that the VPC peering connection, route tables, and security-group rules supported the required private communication between the two application instances.

Test	Expected result	Actual result	Status
Bastion to App-A SSH	SSH connection succeeds	Connection successful	Passed
Bastion to App-B SSH	SSH connection succeeds	Connection successful	Passed
App-A to App-B ping	4 replies, 0% packet loss	4 replies, 0% packet loss	Passed
App-B to App-A ping	4 replies, 0% packet loss	4 replies, 0% packet loss	Passed

All four connectivity tests passed, confirming that the bastion host could access both private instances and that the application instances could communicate in both directions through VPC peering.

9. CONTROLLED FAILURE AND RECOVERY

9.1 Objective:

The objective of this test was to verify how an inbound security group rule affects ICMP communication between two EC2 instances located in different VPCs connected through VPC Peering.

9.2 Procedure:

Step 1: Remove the ICMP Rule

The inbound All ICMP - IPv4 rule was temporarily removed from the security group ewb-sg-app-b. The SSH rule was retained to avoid affecting administrative access.

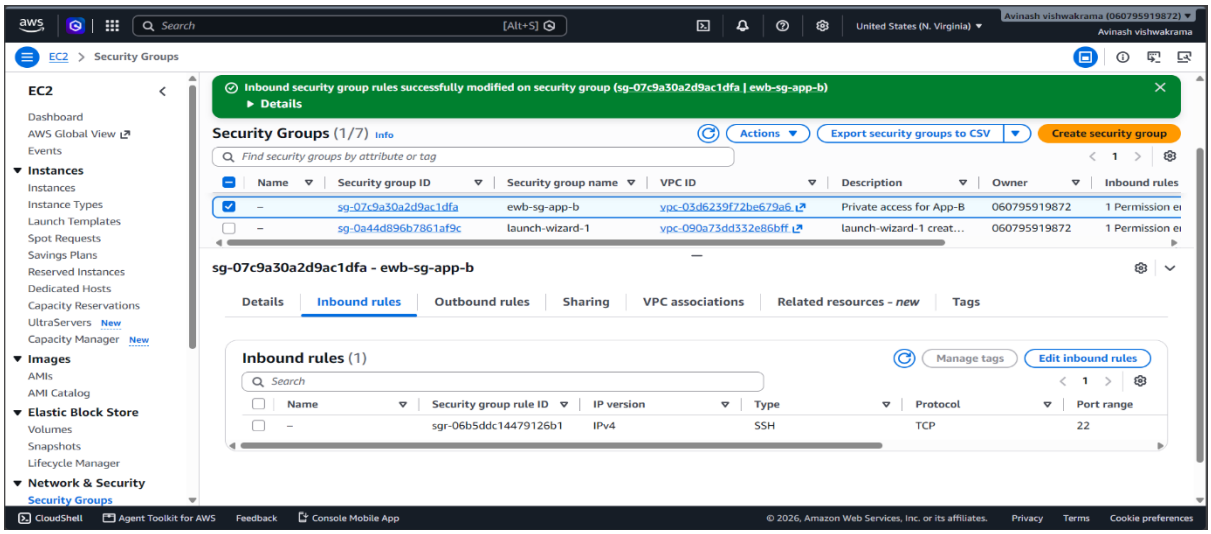


Figure 18 : ICMP rule removed from the App-B security group

Step 2: Verify Connectivity Failure

A ping test was performed from App-A (10.10.2.10) to App-B (10.20.1.10). The test resulted in 100% packet loss because the inbound ICMP traffic was no longer permitted by App-B's security group

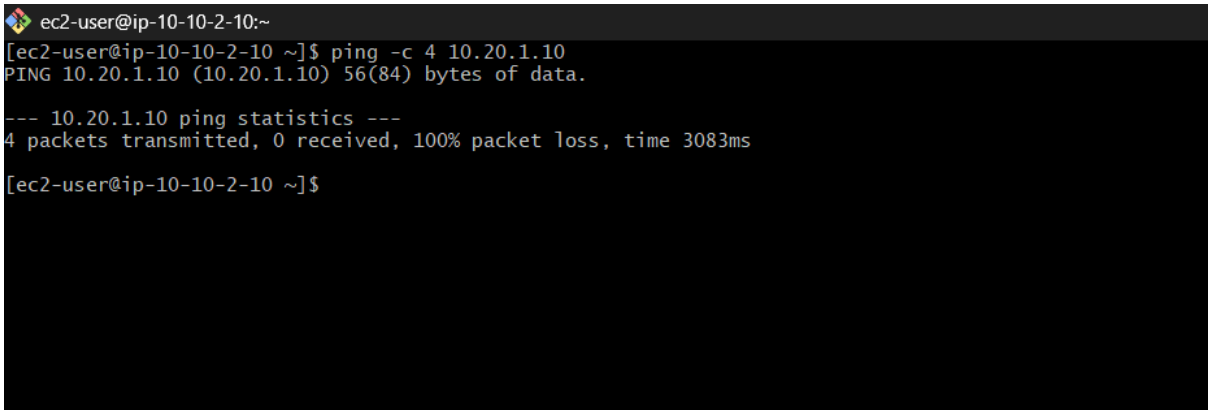


Figure 19 : ICMP connectivity failure from App-A to App-B

Step 3: Restore the ICMP Rule

The inbound All ICMP - IPv4 rule was restored in ewb-sg-app-b, with the source restricted to App-A's private IP address (10.10.2.10/32).

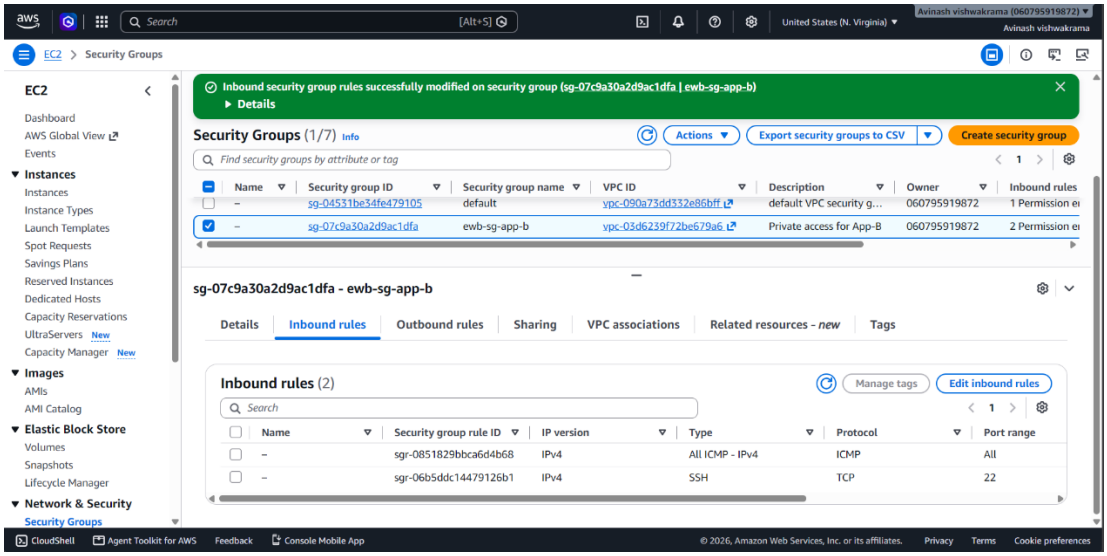


Figure 20 : ICMP rule restored in the App-B security group.

Step 4: Verify Connectivity Recovery

The ping test from App-A to App-B was repeated. All four packets were received successfully, resulting in 0% packet loss.

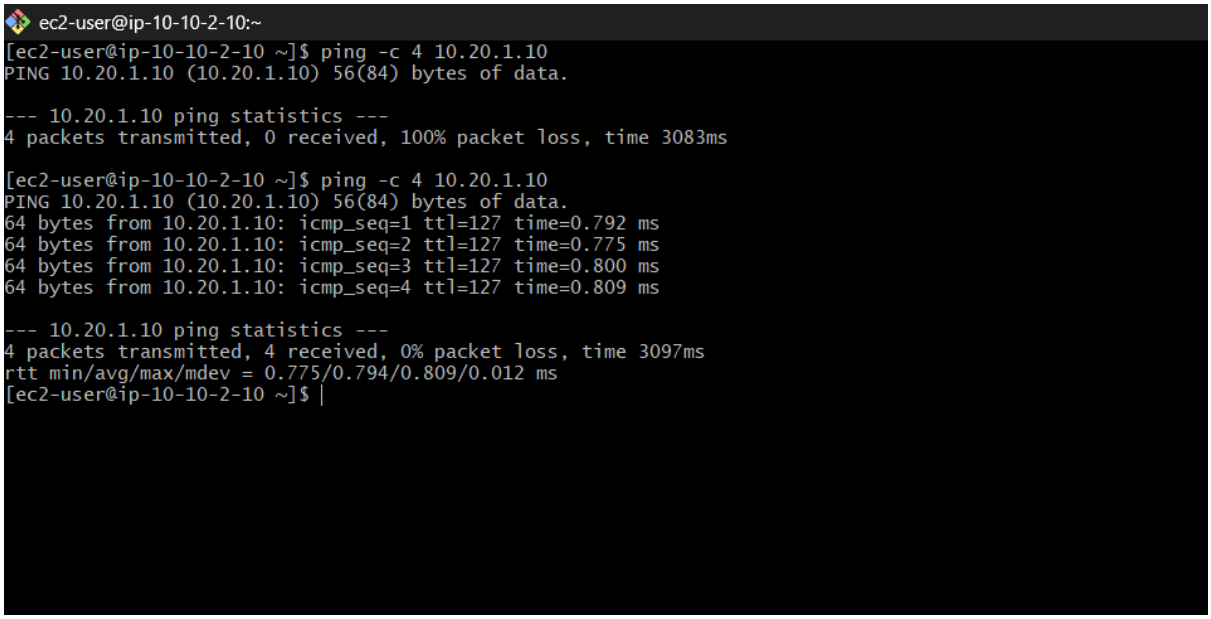


Figure 21 : ICMP connectivity successfully restored from App-A to App-B.

9.3 Result and Observation:

The test demonstrated that ICMP connectivity between App-A and App-B depended on the inbound security group rule configured on App-B. Removing the rule caused the ping test to fail, while restoring it allowed communication to resume. The VPC peering connection and route table configuration remained unchanged throughout the test.

9.4 Conclusion:

The controlled failure and recovery test confirmed that security groups act as traffic filters even when VPC peering and routing are configured correctly. Restricting inbound ICMP traffic to App-A's private IP address also demonstrated the principle of least privilege by allowing only the required source to initiate ICMP communication to App-B.

10. CLEANUP AND RESOURCE DELETION

10.1 Resource Cleanup

After completing the connectivity tests and collecting the required evidence, the AWS resources created for the VPC Peering project were cleaned up to avoid unnecessary ongoing costs.

The cleanup process included terminating the three EC2 instances (ewb-bastion, ewb-app-a, and ewb-app-b), deleting the associated EBS volumes, removing the VPC peering connection, deleting the five project subnets, and deleting VPC B (ewb-vpc-b).

The deletion of VPC A (ewb-vpc-a) was initiated, with its associated Internet Gateway and security groups listed as dependent resources for deletion. The final deletion status was checked in the AWS Console.

10.2 Cleanup Verification

The AWS Console was used to verify resource removal. The EC2 Instances page showed all three instances in the **Terminated** state, the EBS Volumes page showed no remaining volumes in the selected region, the Peering Connections page showed the connection as **Deleted**, and the Subnets page showed no subnets remaining in the selected region.

Screenshots of these states were retained as cleanup evidence. The remaining project resources, including the ewb-lab key pair and any project snapshots, were to be checked separately.

10.3 Learning Note

This project provided practical experience in configuring VPCs, subnets, route tables, security groups, and VPC Peering in AWS. Testing connectivity in both directions helped demonstrate the importance of routing and security group rules for private communication between VPCs. The controlled ICMP failure and recovery test showed how a security group rule can block and restore

connectivity. The cleanup process reinforced the importance of removing unused cloud resources and reviewing billing to control costs.

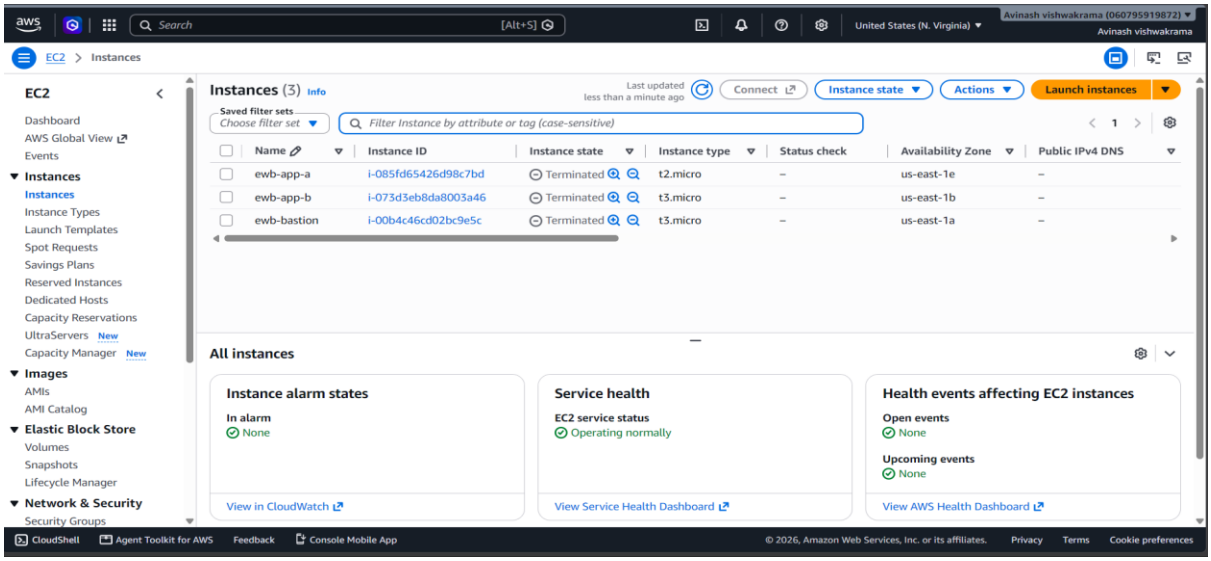


Figure 22: Terminated EC2 Instances

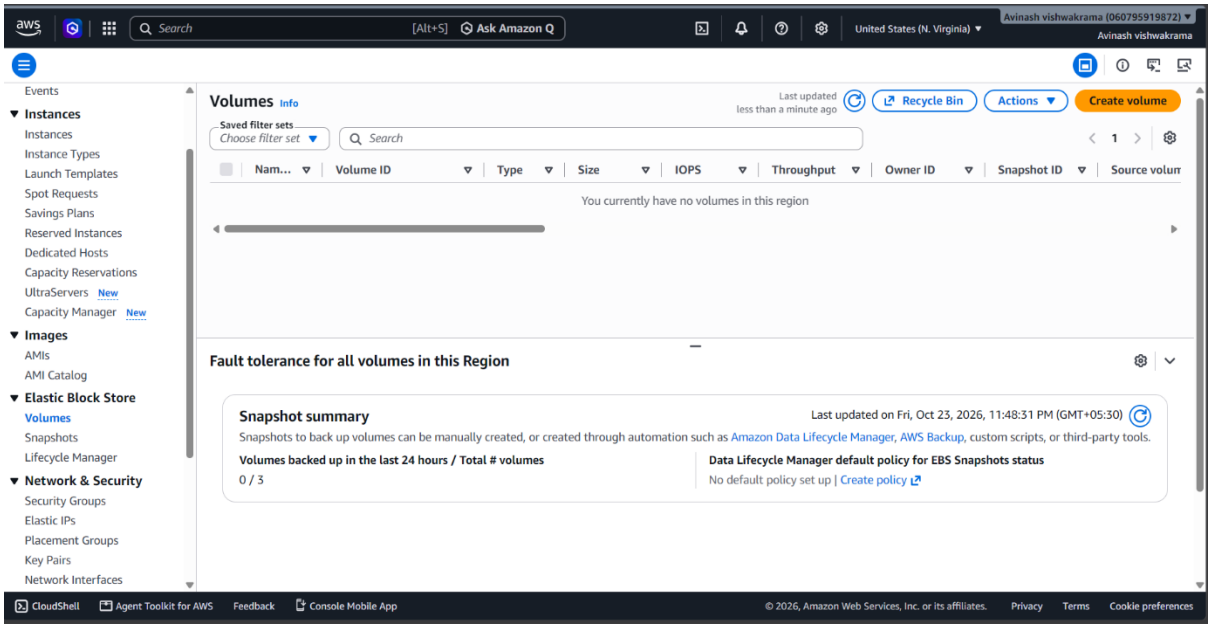


Figure 23: Cleaned-up EBS-VOLUME

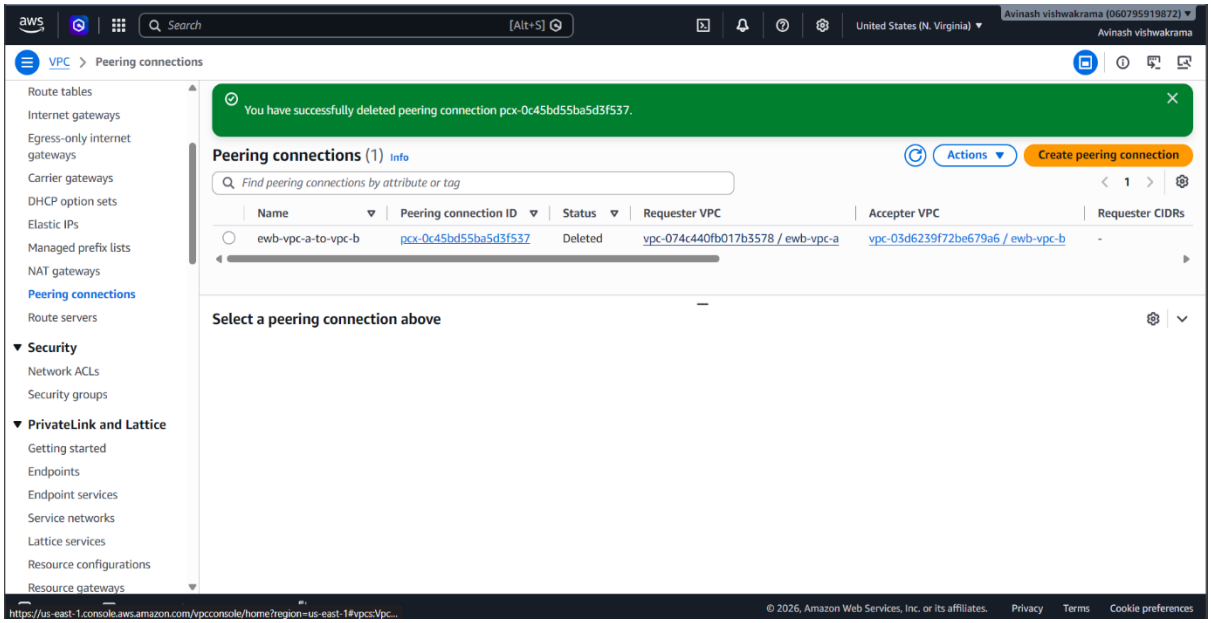


Figure 24: Deleted peering connection

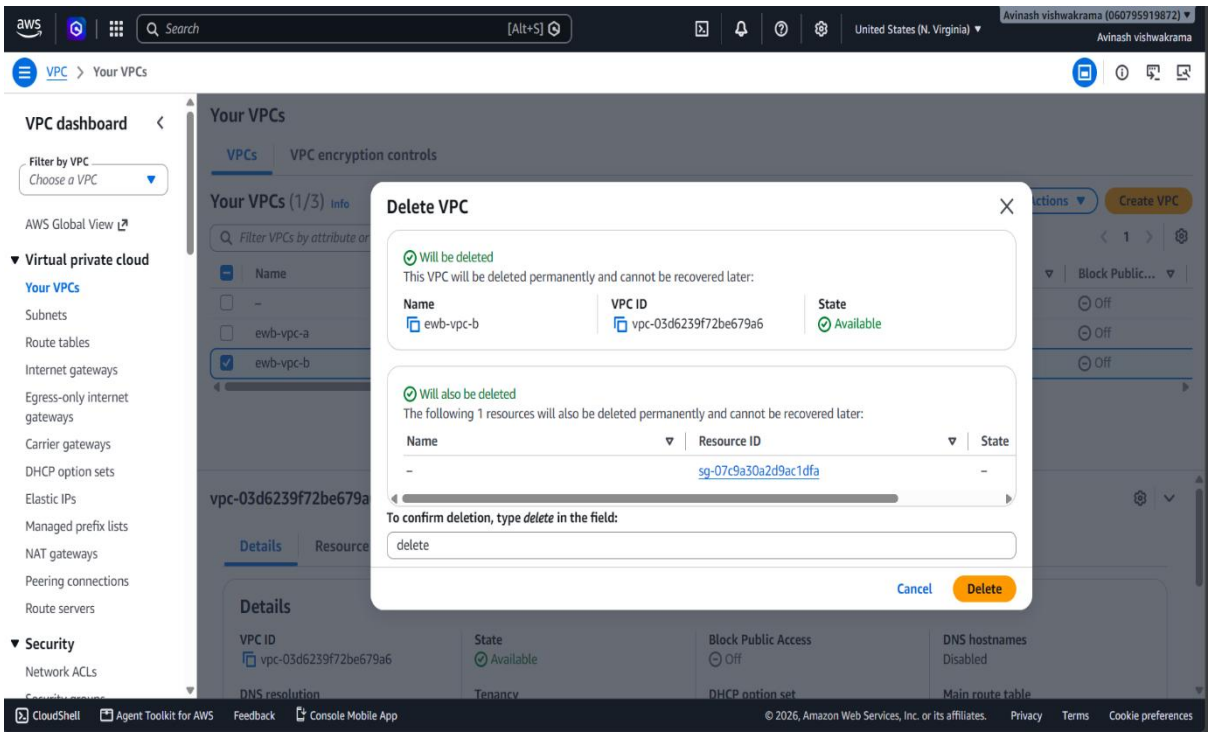


Figure 25: Deleted VPC-b

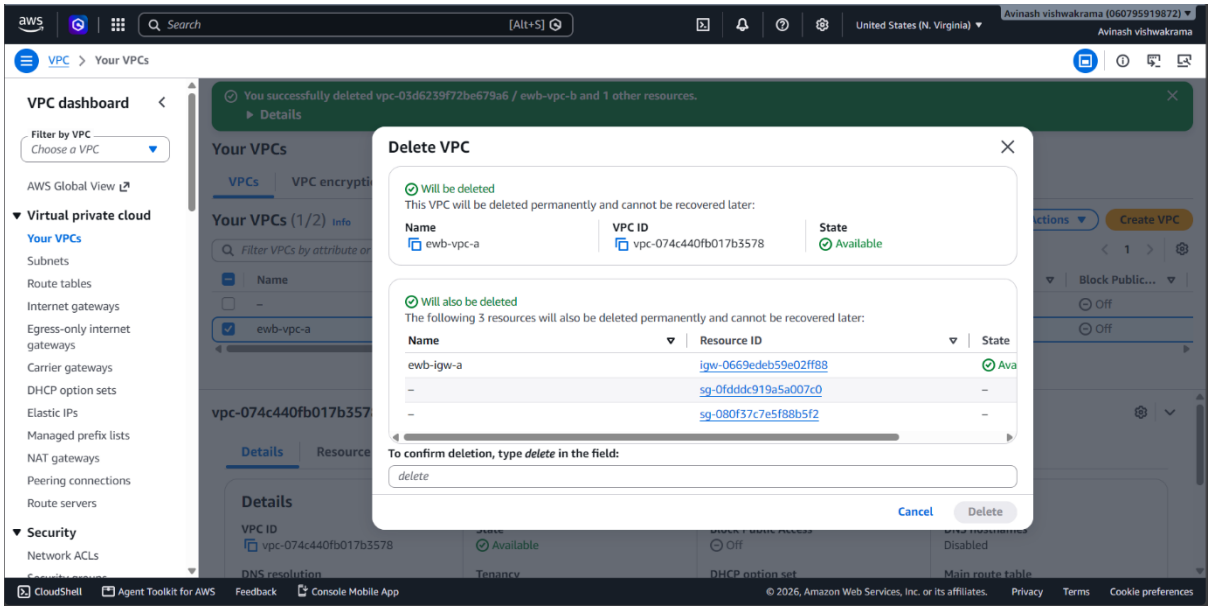


Figure 27: Deleted VPC-a

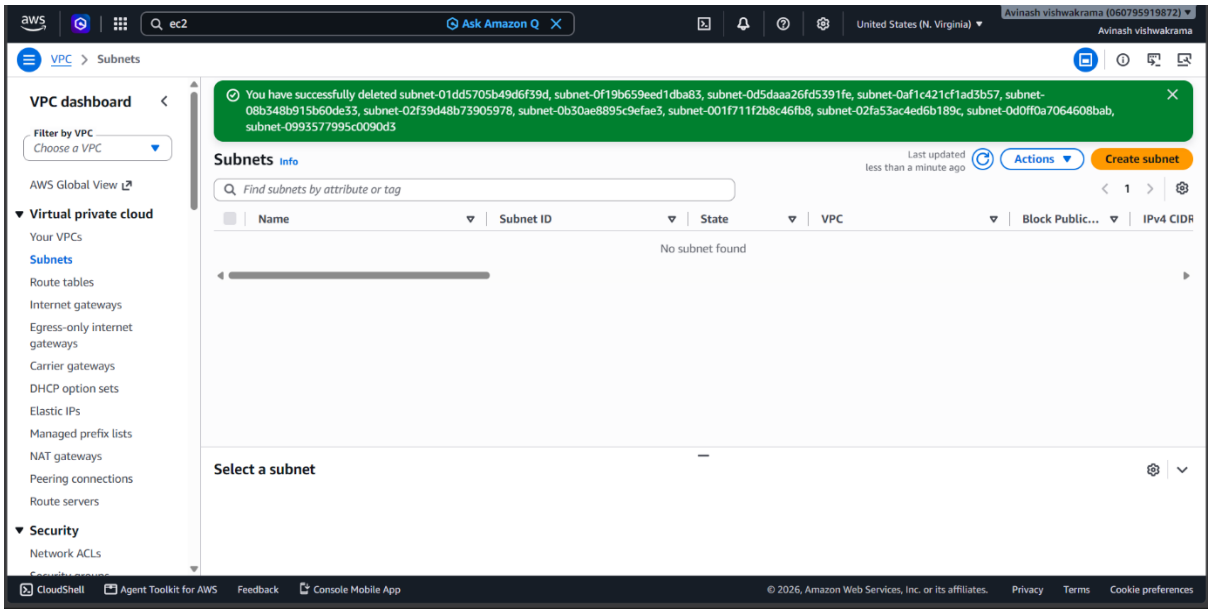


Figure 28: Deleted all associated Subnets

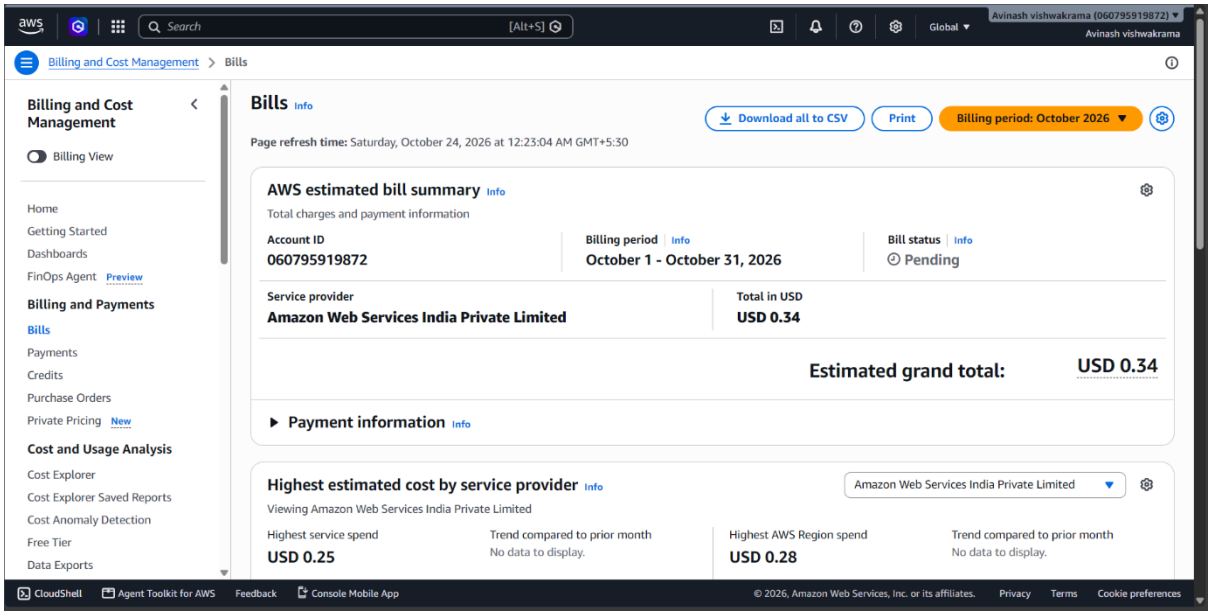


Figure 29: Project Billing

11. LEARNING OUTCOMES AND CONCLUSION

11.1 Learning Outcomes

Through this project, I gained practical experience in the following areas:

- Creating and configuring Virtual Private Clouds (VPCs) with separate IPv4 CIDR blocks.
- Configuring public and private subnets and understanding subnet associations with route tables.
- Establishing VPC Peering between two VPCs and configuring routes for bidirectional communication.
- Configuring security group rules to control SSH and ICMP traffic.
- Connecting to private EC2 instances through a bastion host using SSH.
- Testing connectivity between instances in different VPCs and troubleshooting communication failures.
- Performing a controlled ICMP failure and recovery test to understand the impact of security group rules.
- Cleaning up AWS resources and reviewing billing to help avoid unnecessary costs.

11.2 Conclusion

The project successfully demonstrated how two separate AWS VPCs can communicate privately through VPC Peering when routing and security group rules are configured correctly. Connectivity was verified using SSH and ICMP tests, including a controlled failure and recovery scenario. The project strengthened my understanding of AWS networking, cloud security, troubleshooting, and resource management. It also highlighted the importance of cleaning up unused resources after completing cloud experiments.